

Supply Chain Management

Lecture 1

MM BASC 523

April - May 2026

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**Sauder School Of
Business**

**ROBERT H. LEE
GRADUATE SCHOOL**

SCM

Administration

- **5 Lectures**
- **Lectures**
 - **Lecture Notes will be available on Canvas**
 - Sometimes an incomplete set of slides will be available before lecture.
 - Complete set of slides after the lecture.

Administration

- **Final Exam:**
 - May 19th 9-11
 - Counts for **50%** of the Course Grade.
- **Class Participation (10%)**
 - Please do the readings before class and contribute to the discussion.
 - “Optional”

Administration

Cases

- Two types of Hand ins:
 - Group Cases: Full reports ($7*3 + 10 = 31\%$)
 - Zara+ SHEIN (10), Barilla, Wriston and Intel.
 - Individual Reports (short) (9%)
 - *Merloni* (4%)
 - Dell + HEMA (5%)

Case grading philosophy

- Cases are “**CLOSED**”
- 3 buckets: Above, Met and Below expectations.
 - What they mean
- Expectation on report
- **Analysis Post Case (FEED BACK)**
- **Also Personalized Zoom/in person Feedback**

Final Exam Format

- In person, In class paper based.
- Open notes (to be discussed)

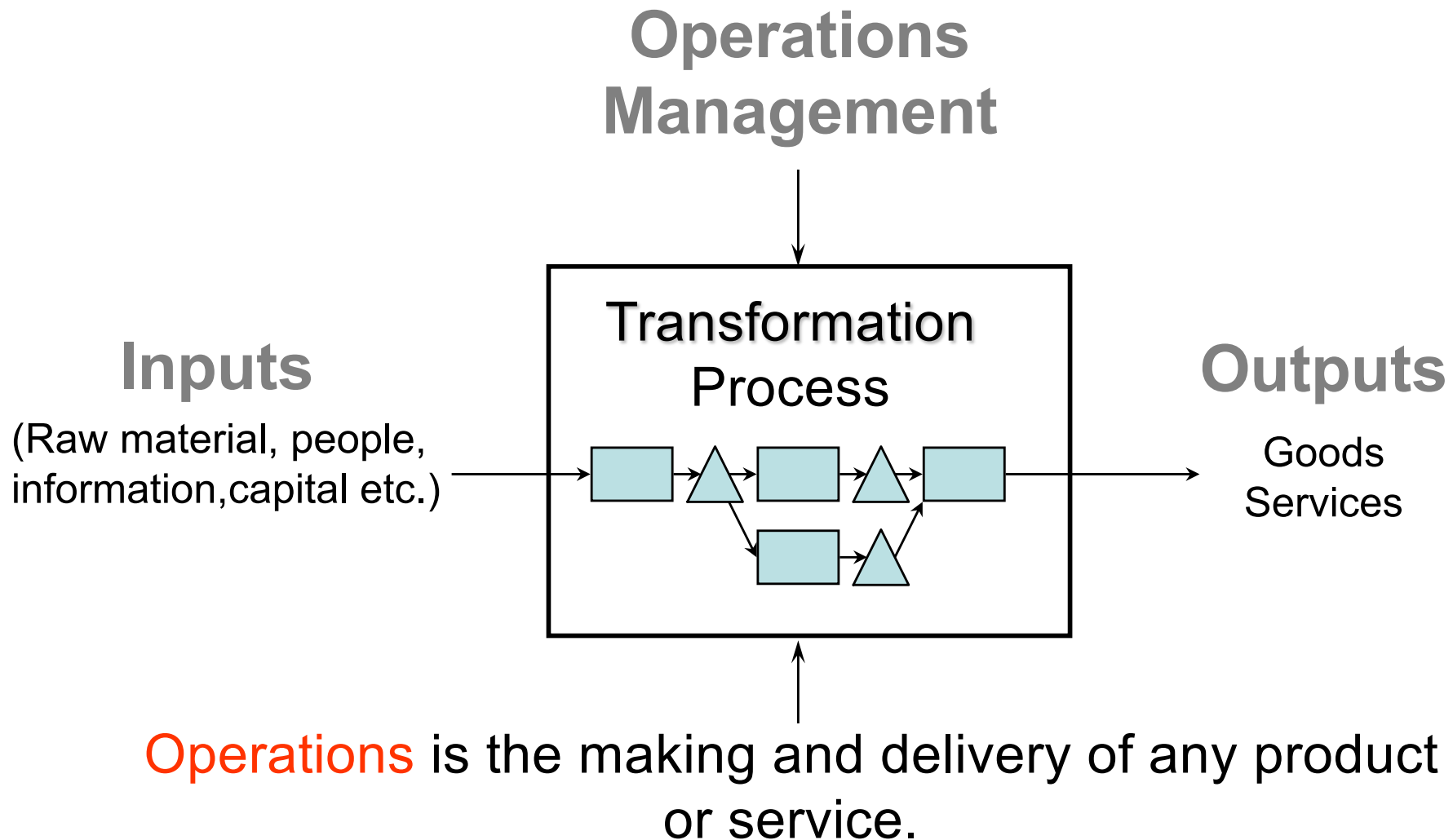
Use of GenAI

- Allowed in specific instances. To be discussed. Please see Case memos.

Today's Agenda

- **Introduction to SCM**
- **Review of Basic Operations.**
- **Dell + HEMA article**
 - Introduces several operational issues/ideas relevant for modern supply chains.
- **Supply chain Models and Structure**

What is Operations Management (OM)?



Understand key principles

Framework

Processes

Inventory

Variability

Capacity

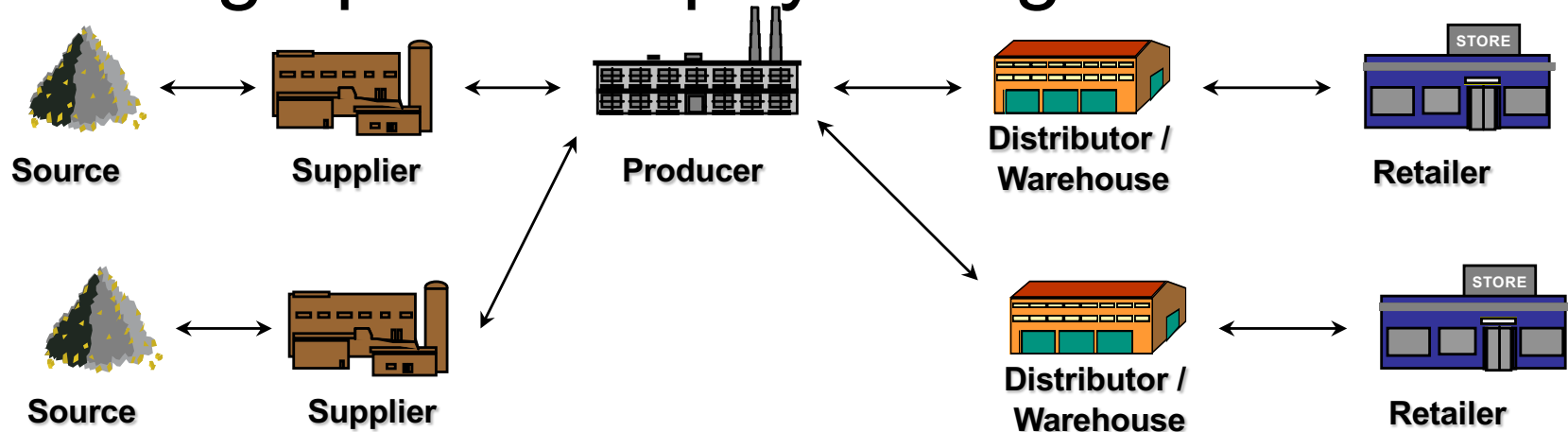
Measurement

Complexity

Moving from Processes to Supply Chain Management

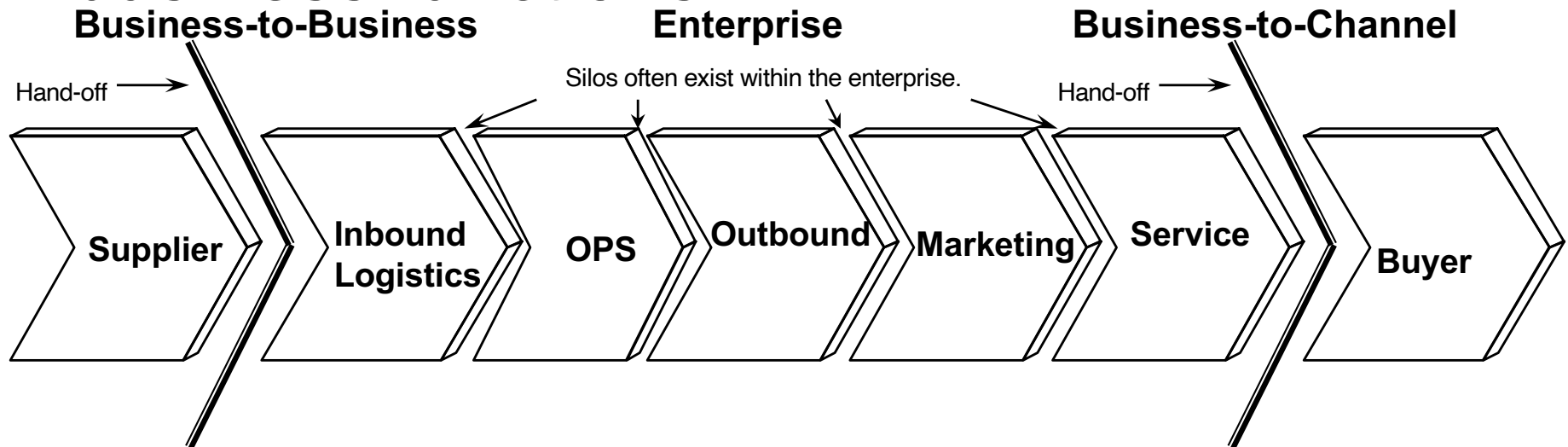
What Is a Supply Chain?

- A supply chain is a network of flows and storage points of physical goods

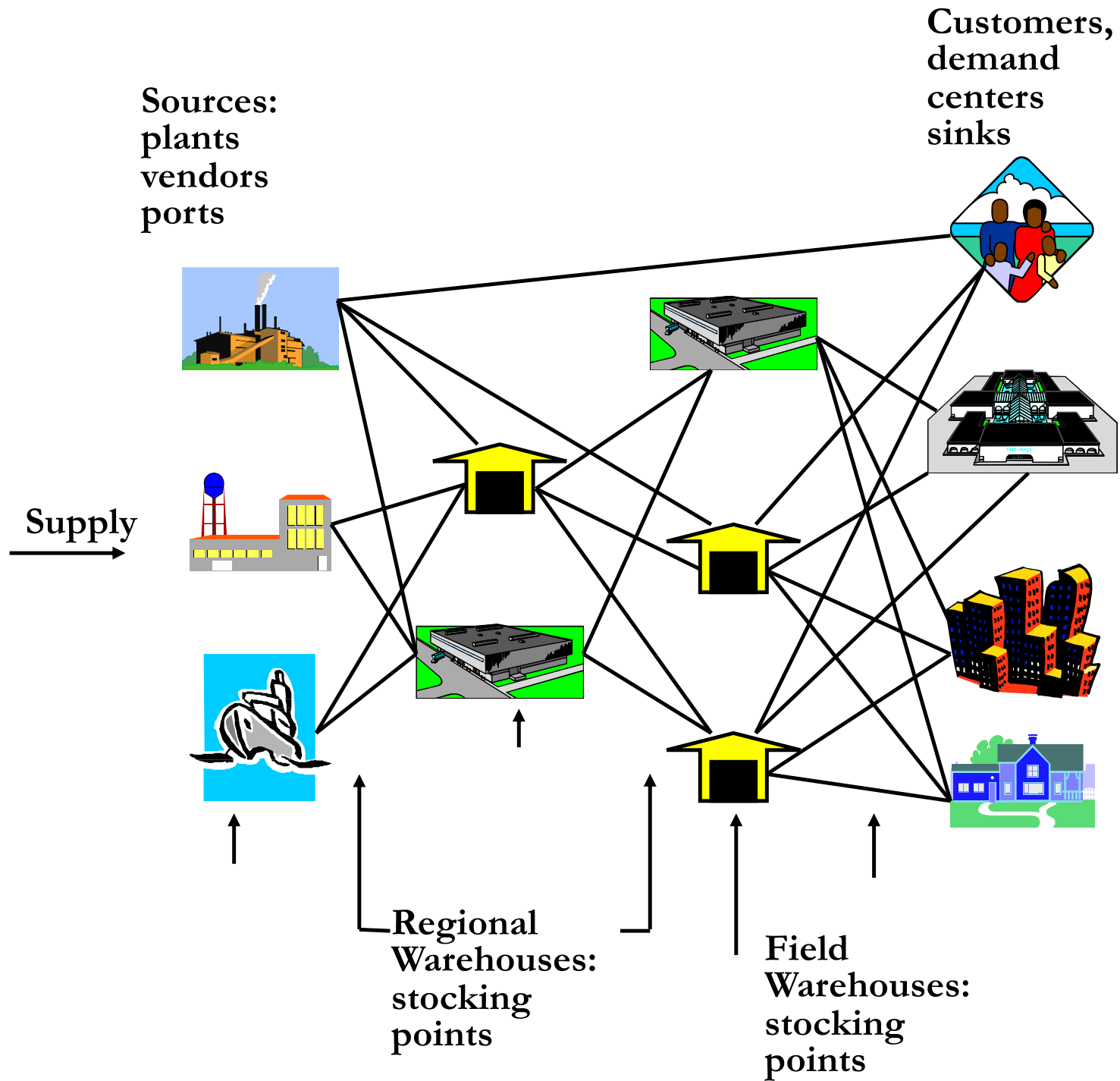


What Is a Supply Chain?

- A supply chain is an integrated set of business functions



Sourcing, purchasing, forecasting, production planning, order processing/tracking, accounting, transportation planning, warehousing, sales, customer support and service, and more.



- Managing a supply chain is very challenging...



2011 Great Sendai earthquake and Tsunami

- Lives Lost
- Injuries
- Property, business...(400,000 Cars)
- Huge economic Loss
- Massive Supply Chain Disruption lasting many years

Supply chain structure

- **Car Manufacturers**
 - 5000 Tier 1 supplier, each with an average of 250 tier 2 suppliers
- **938 of Fortune 1000 companies have at least two or greater Tier 1 or Tier 2 suppliers in Wuhan!!!**

Managing a supply Chain

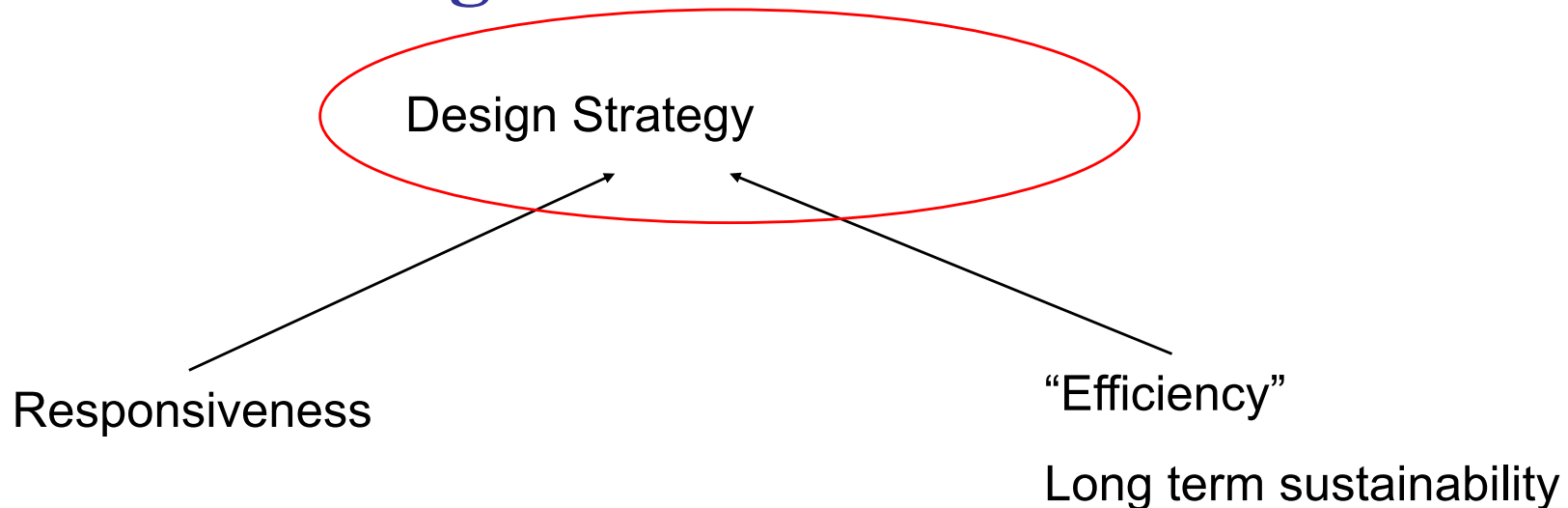
- **CoVisInt**
 - Collaboration
 - Visibility
 - Integration

Supply Chain Design

- A few different frameworks...

Factors That affect design

- **Customer needs that are met**
- **Cost meeting these needs**



Factors That affect design

- **Customer needs that are met**

-

-

-

- **Cost meeting these needs**

-

-

-

The Six Supply Chain Drivers

- **Sourcing**
- **Pricing**
- **Inventory**
- **Transportation**
- **Facilities & Design**
- **Information**

The Six Supply Chain Drivers

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The Six Supply Chain Drivers

- Sourcing
- Pricing
- Inventory
- Transportation
- Facilities & Design
- Information

**Role, location, capacity,
flexibility**

Metrics

Capacity, utilization, cycle times, variety, Volume contribution, set ups, batch sizes, service levels...

The Six Supply Chain Drivers

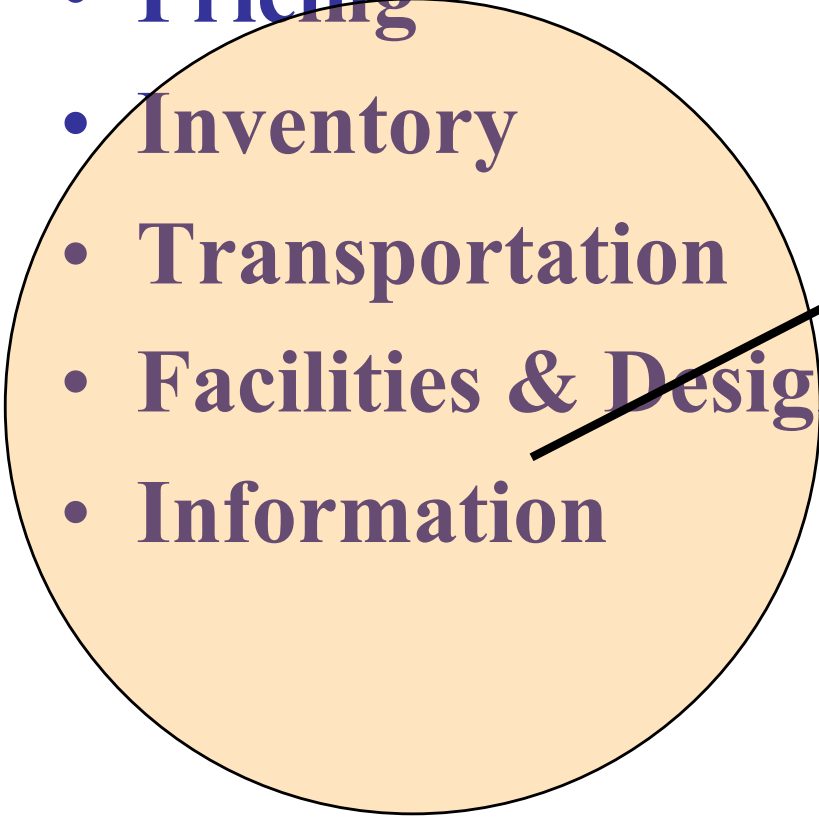
- Sourcing
- Pricing
- Inventory
- Transportation
- Facilities & Design
- Information

Raw materials, work in process, finished goods...

Metrics

Average Inv., Days of inventory, replenishment size, safety stock, fill rates, time out of stock measures...

The Six Supply Chain Drivers

- Sourcing
 - Pricing
 - Inventory
 - Transportation
 - Facilities & Design
 - Information
- 

**Demand and customer info.,
information to customer on
orders, availability...**

**Potentially the biggest driver
of performance**

Metrics

Forecast horizon, error, degree
of order transparency, variance
from plan, ratios --- order to
demand variability...

The Six Supply Chain Drivers

- Sourcing
- Pricing
- Inventory
- Transportation
- Facilities & Design
- Information

**Modes, routes, packing,
mixing etc...**

Metrics

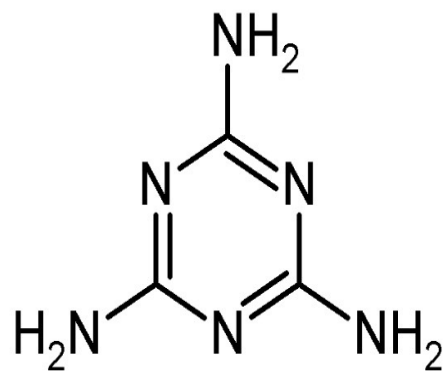
Inbound costs, sizes, outbound
costs and sizes, fraction
transported by mode

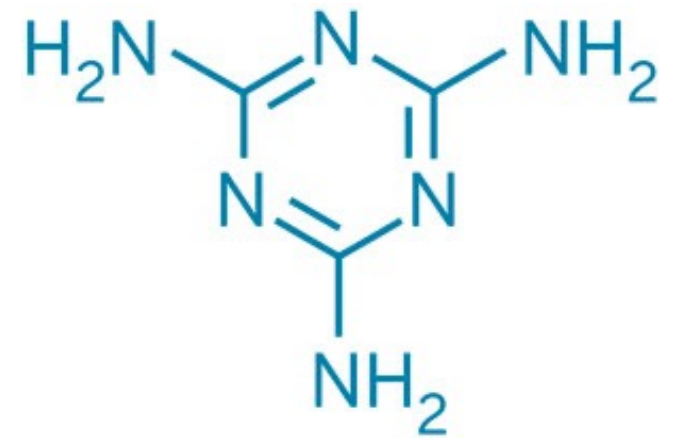
Outcome driven supply chains

Factors in designing supply chains:

- Cost
- Responsiveness
- Security
- **Sustainability**
- Resilience
- Innovation

What is melamine?





Melamine

300,000 illness. Including children
Deaths
9000 Tons of milk recalled
Execution and Imprisonment of individuals.

Details on melamine in milk

- September 2008: Reports of 14 infants in Gansu province, China, ill with renal disease after drinking formula made from same formula powder
- Cause determined to be melamine based on characteristic crystals in kidneys
- Of 491 formula batches tested, 69 samples from 22 companies were positive; concentrations ranged from 0.09 to 2563 mg/kg
 - Codex Alimentarius Commission (WHO) limits are 1 mg/kg for infant formula, 2.5 mg/kg for other foods
- Melamine likely added by milk dealers/processors, not farmers

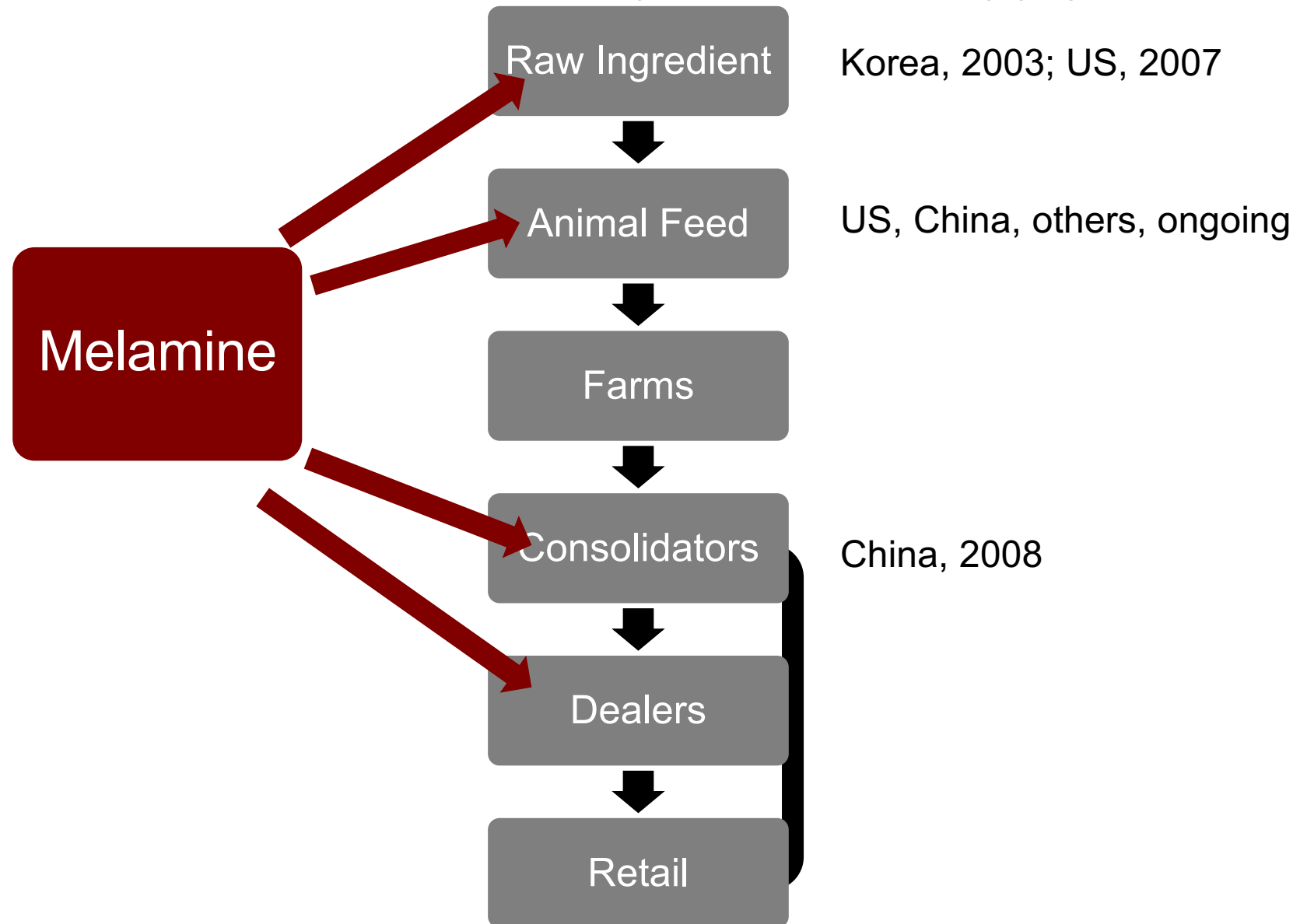
Consequences of melamine in milk

- **Melamine discovered in several related products (e.g., candy, yogurt)**

China and USA

- Testing for Melamine
 - How?
 - When?
 - What?

Melamine Points of Entry in Animal Supply Chains



Preliminary Ingredient Map

Adulterant

Ingredient

Final Product

melamine

wheat gluten

milk

animal feed

pet food

meat substitutes

bread

beer

sauces

ice cream

infant formula

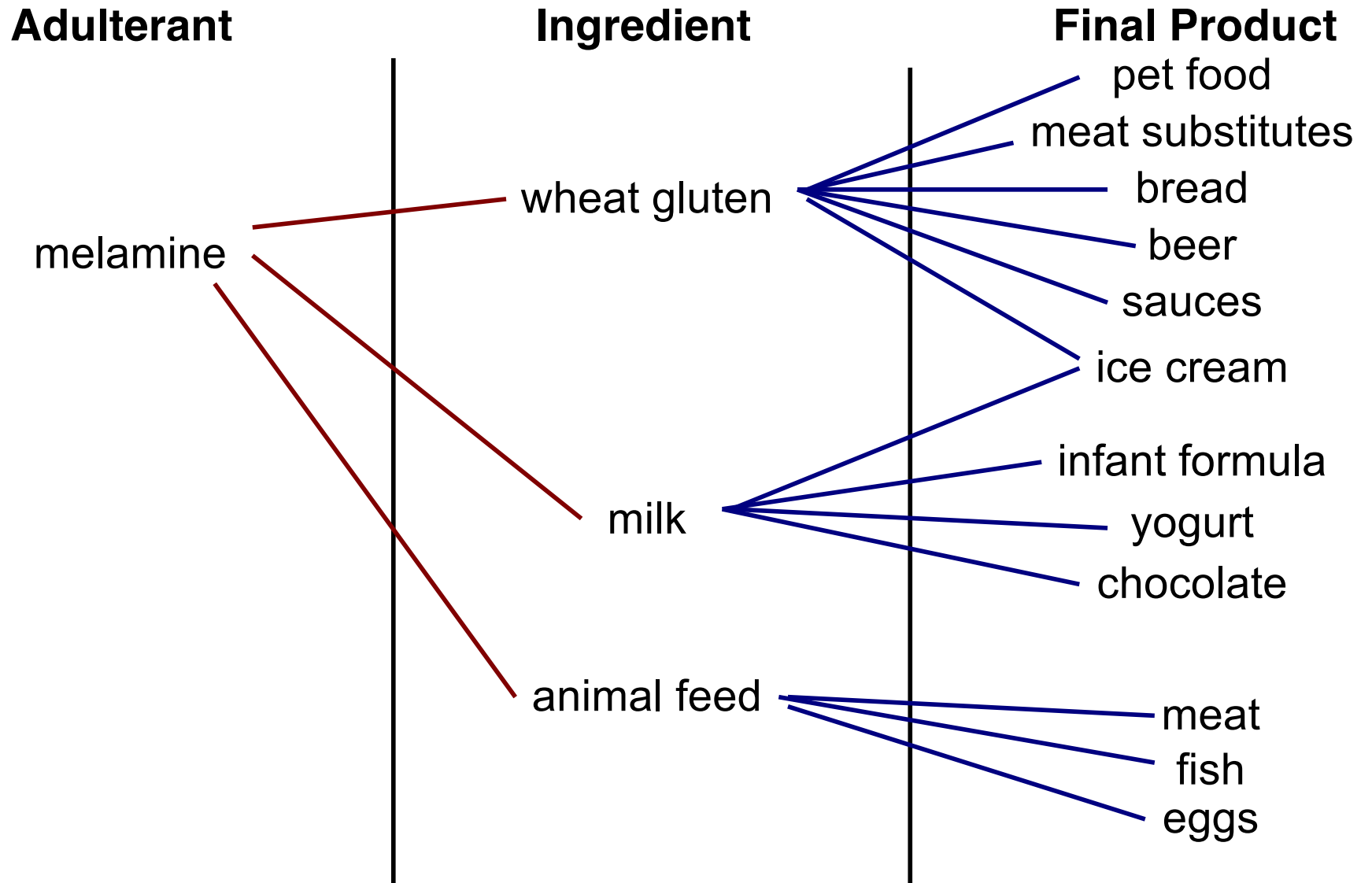
yogurt

chocolate

meat

fish

eggs

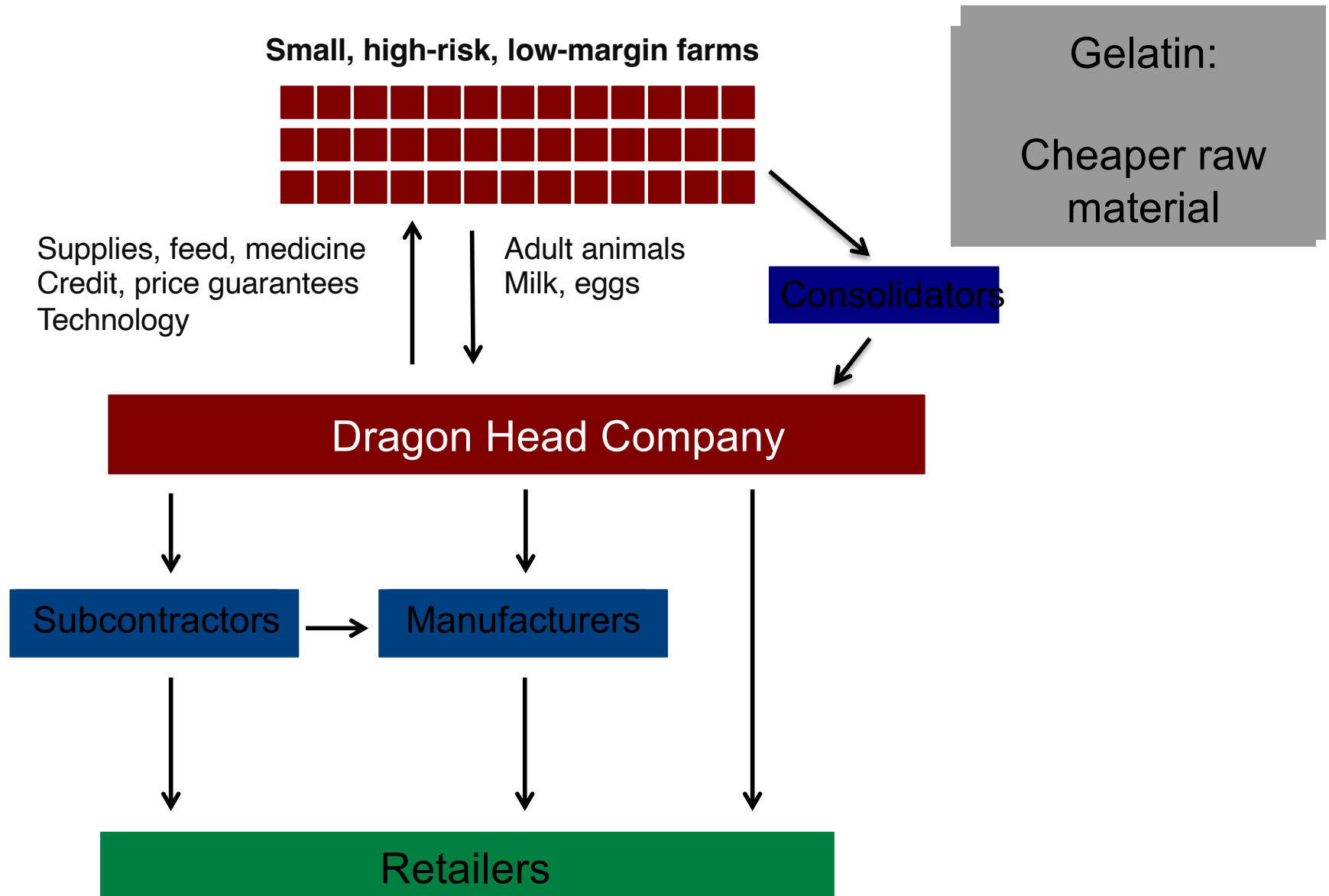


Why is this important?

Risk Management Challenges

- Globalization of supply chains particularly from emerging markets
- FDA-Regulated imports grow at a rate of 13% annually.
- Increase in online purchases of drugs, supplements and food products
- Socio-economics pressures motivate counterfeit and adulterations
- Traditional regulatory enforcement approaches (site & port inspections) are insufficient and lag behind

Dragon Head Farming



Risk Drivers

- Animal-based supply chains with dragon-head farming structure
 - Chicken, Honey, Milk, Heparin, Gelatin
 - Seafood and fish?
- **Dispersion and traceability**
- Implementation of new regulations
 - Taxes (honey), Regulatory effort (milk), Economic regulations (gelatin)

Supply Chain Strategy at and Dell & HEMA

Dell Operations

Evolution of the PC Industry

Computers

(By shipment, market share)

Top 6 vendors by number of units shipped, 2025			
Rank	Manufacturer	Country of origin	Market share ^[1]
1	Lenovo	China	27.2%
2	HP	USA	21.3%
3	Dell	USA	15.3%
4	Apple	USA	9.2%
5	Asus	Taiwan	6.9%
6	Acer	Taiwan	6.3%
-	other	various	13.9%

PC Industry 2004

	Dell	IBM	Apple	HP
Revenue (billion \$)	49.2	96.3	8.3	79.9
Net income (billion \$)	3.0	8.4	0.3	3.5
Number of employees	55,000	329,000	13,400	151,000
Revenue per employee	\$ 900,000	\$ 300,000	\$ 620,000	\$ 530,000
Days of inventory	4	22	6	44

Source: COMPUSTAT database

Dell Inventory Trends...

• Year	Inventory Turnover		Days of Inv.
1992	4.79		76
1993	5.16	Reported Loss	70
1994	9.4	Withdrew from Retail	38
1995	9.8		---
1996	24.2	Dell.com	15
1997	41.7		9
1998	52.40		7
1999	52.40		---
2000	51.4		---
2001	63.50		5

Most efficiency measures have
limitations!

Retail Industry Segment	Inventory Turnover	Gross Margin
Apparel And Accessory Stores	4.57	37%
Catalog, Mail-Order Houses	8.60	39%
Department Stores	3.87	34%
Drug & Proprietary Stores	5.26	28%
Food Stores	10.78	26%
Hobby, Toy, And Game Shops	2.99	35%
Home Furniture & Equip Stores	5.44	40%
Jewelry Stores	1.68	42%
Radio,TV, Cons Electr Stores	4.10	31%
Variety Stores	4.45	29%

Direct Business Model (DBM)

- **For today's discussion, let us assume that DBM means a manufacturer or producer of a product or service directly selling to customers.**

Direct Business Model

- Manufacturers sells directly to customers

“distance between creator and consumer is very small”

selling *Tide* on-line?

CONSUMER PRODUCTS



P&G jumping into online retail with new test site

4/11



Direct Business Model

- Manufacturers sells directly to customers
“distance between creator and consumer is very small”

What are some ADVANTAGES of this model?

DBM: Advantages

- Better Demand Information
- No “Middleman”: Therefore higher Margins
 - Called double marginalization
- Ability to customize
 - Higher Variety
 - Increases product’s attractiveness
 - Higher Market Share
 - Extracts consumer’s “Willingness to Pay”

Direct Model

- Variability Reduction (Better information)
- Customization

Online Grocery Industry

- **Vancouver**

- **Global**

- **China**

– **Landscape?**

Models

1. Ship from Retail (Store + Warehouse)

– HEMA, 7Fresh, Ella (dead)

2. Ship from central Warehouse

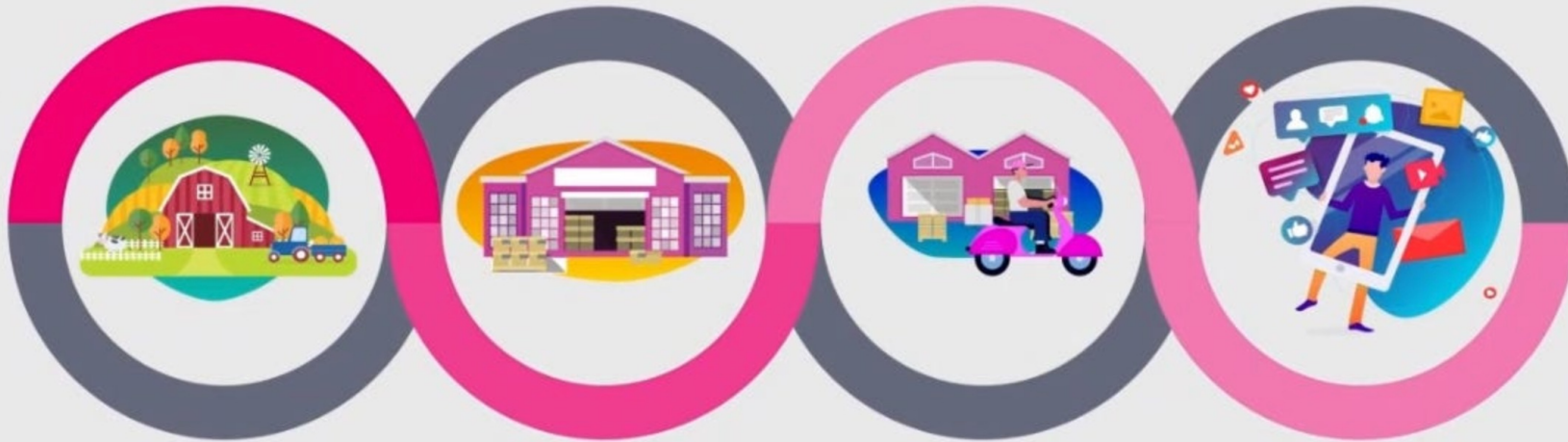
3. Front end warehouse (DMW model)

– Dingdong, Meituan, MissFresh (dead?)...

	Front-end Warehouse	Store + Warehouse
Introduction	Set up warehouses for sorting and distribution close to the consumer, shortening the distribution chain to reduce costs.	Instore catering + online shopping + instant delivery or pick-up. Provides online and offline integrated consumption experience.
Cities	Mainly 1st & 2nd-tier cities	Mainly 1st & 2nd-tier cities
Delivery range	1-3 km	1-5 km
Delivery time	30-60 min	30-60 min
Major companies	MissFresh, Dingdong Maicai, Pupu, Meituan Maicai	Hema, 7Fresh

**300km radius coverage
Processing & QC Centers**

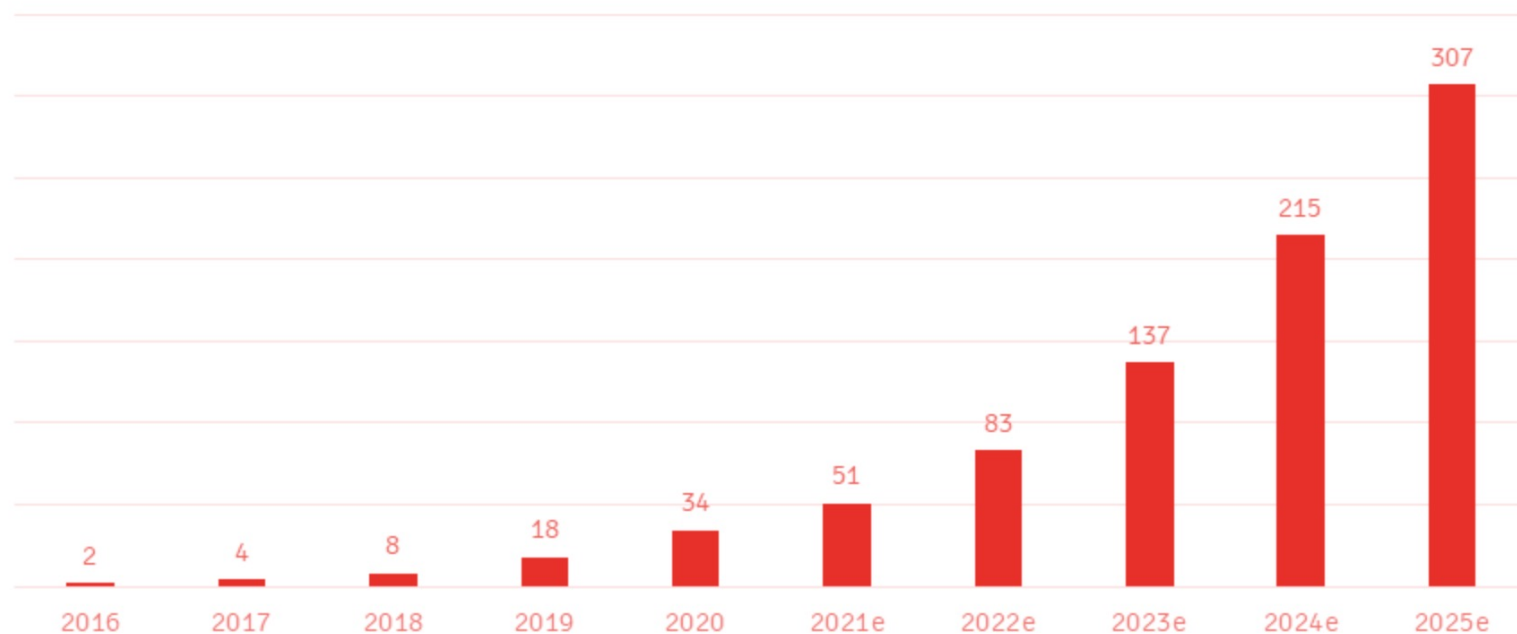
Customers



Origin

**3km radius
community coverage DMWs**

GMV Front-end Warehouse Market (billion RMB)



Source: iResearch

Discussion

- Operating Model and Strategy
 - Advantages and disadvantages

Dell and Customization

- Mass customization

Customization / Variety

- Can you think of Large firms that sell you a customized product?

Customization / Variety

- Can you think of Large firms that sell you a customized product?
- Why is customization Hard?

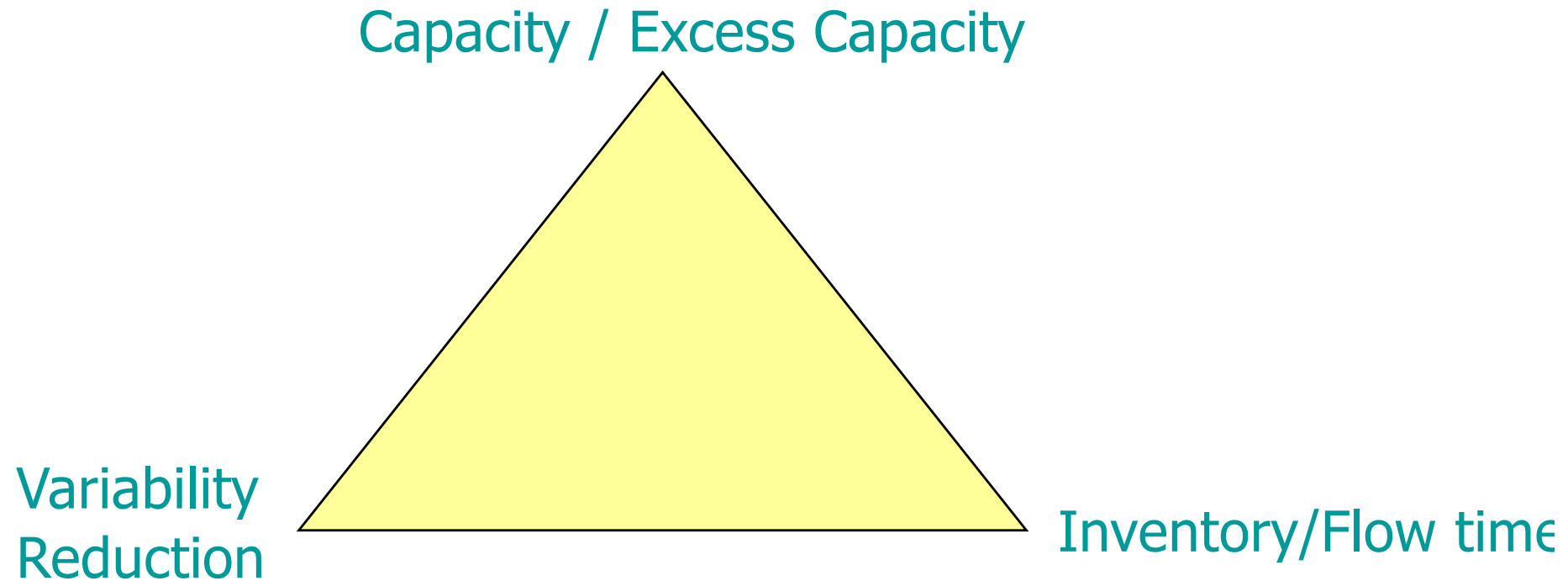
Ideas!

- **Delayed Differentiation**
 - Postpone the step that adds variety to the end of the process.

Ideas!

- **Delayed Differentiation**
 - Postpone the step that adds variety to the end of the process.
- **Standardized components**
 - *“Design to manufacture”*
- **Flexible tooling**
- **Variability reduction**

OM Triangle



Capacity, inventory and variability reduction (information) are substitutes in running operations...